

II-3 Superprompt-Critical Excursion Following a Step Reactivity Insertion¹

Investigation of the Differential Equation

Superprompt-critical transients induced by a reactivity step with feedback described by the linear energy model can be treated analytically if the prompt kinetics approximation is applied, thus providing an important semi-quantitative understanding of the power burst.

The differential equation and the initial condition in the prompt kinetics approximation are given by

$$\dot{p} = \frac{\rho - \beta}{\Lambda} p , \quad (\text{II.74})$$

with

$$p(0) = p^0 = \frac{\rho_1}{\rho_1 - \beta} p_0 . \quad (\text{II.75})$$

The reactivity contains the Doppler feedback as described by an energy coefficient γ , Eqs. (II.38). Again, γ is assumed to be independent of temperature so that simple theoretical relations between the kinetics and feedback parameters can be derived. The temperature rise is calculated on the basis of the assumption of adiabatic boundary conditions (linear energy model):

$$\rho(t) = \rho_1 + \gamma \int_0^t p(t') dt' . \quad (\text{II.76})$$

The stationary cooling is neglected, i.e., the term “ $-p_0$ ” is omitted in the integrand of Eq. (II.76). This neglect is physically not well justified in most transients, but it greatly simplifies the mathematical treatment. The stationary cooling during a transient is equivalent to a positive ramp reactivity insertion, i.e.,

$$\gamma \int_0^t (-p_0) dt' = -\gamma p_0 t . \quad (\text{II.77})$$

Thus, simple formulas describing the *step-induced* excursion cannot be obtained without neglecting the stationary cooling. The stationary cooling is included in the investigation of ramp-induced transients, which are discussed in the next section.

For convenience, the prompt reactivity $(\rho - \beta)$ is designated by a single quantity ρ_p , standing for the reactivity above β :

$$\rho - \beta = \rho_p . \quad (\text{II.78})$$

The prompt kinetics equation then assumes the form

$$\dot{p} = \frac{\rho_p}{\Lambda} p , \quad (\text{II.79})$$

with

$$p(0) = p^0 = \frac{\rho_1}{\rho_{p1}} p_0 . \quad (\text{II.80})$$

¹ Source:

1. Ott & Neuhold, *Introductory Nuclear Reactor Dynamics*, American Nuclear Society, 1985

After the onset of the transient, the flux rises rapidly or promptly. The corresponding flux integral causes, through the negative feedback, the reactivity to decrease. At $t = t_m$, the decreasing reactivity becomes equal to β , i.e., $\rho_p = 0$. Then the derivative in Eq. (II.79) becomes equal to zero. It follows that the prompt reactivity vanishes at the same time when the flux has a maximum:

$$\rho_p(t_m) = 0 \quad (\text{II.81a})$$

or

$$\rho(t_m) = \beta. \quad (\text{II.81b})$$

Substituting into Eq. (II-76) gives

$$-\gamma \int_0^{t_m} p(t') dt' = \rho_1 - \beta. \quad (\text{II.82})$$

Consequently, *the prompt part of the inserted reactivity is just compensated when the flux has reached its maximum.*

Equation (II.82) provides directly the energy release during the first part of the transient, i.e., from the beginning to the maximum of the excursion (with γ and p replaced by energy coefficient and power, γ_e and P):

$$\Delta Q_m = \int_0^{t_m} P(t') dt' = -\frac{\rho_{p1}}{\gamma_e}. \quad (\text{II.83})$$

Thus, the energy release during the rising part of the transient is given by the ratio of the prompt reactivity inserted and the energy coefficient. Note the important fact that the *energy release* ΔQ_m is independent of the neutron generation time.

Equation (II-83) as well as the corresponding equation for the temperature rise may be rewritten as difference quotients. They then become special cases of the corresponding feedback coefficients:

$$\frac{\Delta \rho}{\Delta Q} = \frac{\partial \rho}{\partial Q} = \gamma_e = \text{energy coefficient} \quad (\text{II.84a})$$

and

$$\frac{\Delta \rho}{\Delta T} = \frac{\partial \rho}{\partial T} = \gamma_T = \text{temperature coefficient}.$$

Equations (II.84a) hold at all times. At the maximum of the transient, one has $\Delta \rho = -\rho_{p1}$. Then Eqs. (II.84a) can be solved for ΔQ_m and ΔT_m :

$$\Delta Q_m = -\frac{\rho_{p1}}{\gamma_e}, \quad (\text{II.84b})$$

and

$$\Delta T_m = -\rho_{p1} \left(\frac{\partial \rho}{\partial T} \right)^{-1}.$$

Inserting the typical feedback coefficient of Eqs. (II.46) and (II.47), the following values are obtained for $\rho_1 = 1.1\%$ as an example:

$$\Delta Q_m \approx \frac{0.1}{0.8} P_n \text{ seconds} \approx 300 \text{ MW} \cdot \text{s} \quad (\text{for } P_n = 2500 \text{ MW}) \quad (\text{II-85a})$$

and

$$\Delta T_m \approx 5 \text{ K (for LWRs)} \quad \text{or} \quad \Delta T_m \approx 50 \text{ K (for FBRs)} .$$

The *rate* of the reactivity reduction by feedback is largest at the power maximum:

$$\dot{\rho}_m = \gamma_e P_m . \quad (\text{II-86})$$

Thus the reactivity quickly passes through the value of β and into the range below prompt critical.

The applicability of the prompt kinetics approximation (PKA) below $\rho = \beta$ is questionable. During a prompt jump or in cases in which the delayed neutron source practically determines the flux level through source multiplication, the application of the PKA gives unrealistic results since it cannot account for the behavior of the delayed neutron source, except in the initial condition. Below $\rho = \beta$, the prompt kinetics model only describes the prompt die-away of an existing neutron population. Since it is just the die-away of the large neutron population built up before the maximum of such transients, the PKA yields quite accurate results during this phase of the transient. During this phase, the flux level is reduced from its maximum value to a level determined by source multiplication of the delayed neutrons. After this phase, however, the PKA is not applicable.

In terms of the microkinetics, the superprompt transient consists of buildup of diverging fission chains. With ρ turned back below β through feedback, these chains cease to diverge and begin to die away. The die-away is described by the prompt period, which is the quantity appearing as the decay constant in the prompt kinetics equation, Eq. (II-74).

In conclusion, it is emphasized that the applicability of the PKA model is normally restricted to the reactivity range above β . Below β , it always yields a die-away of the neutron population that is normally incorrect. But this allows the exceptional applicability of the PKA to part of the range $\rho < \beta$ if $\rho(t)$ enters this range quickly from the superprompt domain because then the ensuing transient consists merely of the die-away describable by the PKA until finally the delayed neutrons take over. The quantitative comparisons below illustrate these aspects in detail.

Investigation of the First Integral

The consideration of the differential equation for superprompt-critical transients with linear energy feedback, Eq. (II-74), has already provided important information about the reactivity, energy release, and temperature rise between the onset and the maximum of the excursion. Further information is obtained by considering the "first integral" of the original equation, which can be brought in the form of a differential equation of second order.

Equations (II-74) and (II-76) are rewritten into a second-order differential equation. By denoting the integral in Eq. (II-76) with q , which gives $\dot{q}$ for p and $\ddot{q}$ for $\dot{p}$, one obtains instead of Eq. (II-74):

$$\Lambda \ddot{q} = (\rho_{p1} + \gamma q) \dot{q} \quad (\text{II-87})$$

and

$$q(t) = \int_0^t p(t') dt' .$$

An even simpler equation is obtained by introducing the prompt reactivity as the unknown function. The reactivity ρ_p and p are related as [see Eqs. (II-76) and (II-78)]:

$$\dot{\rho}_p = \gamma p = \gamma \dot{q} \quad (\text{II-88a})$$

and

$$\ddot{\rho}_p = \gamma \dot{p} = \gamma \ddot{q}. \quad (\text{II-88b})$$

Multiplying Eq. (II-79) with $\Lambda\gamma$ and inserting Eqs. (II-88) yields:

$$\Lambda \ddot{\rho}_p = \rho_p \dot{\rho}_p, \quad (\text{II-89})$$

i.e., a nonlinear second-order differential equation.

The first integral of Eq. (II-89) is readily found:

$$\Lambda \int_0^t \frac{d\dot{\rho}_p}{dt'} dt' = \int_0^t \rho_p \frac{d\rho_p}{dt'} dt', \quad (\text{II-90})$$

which gives a first integral in the form

$$\Lambda [\dot{\rho}_p(t) - \dot{\rho}_p(0)] = \frac{1}{2} [\rho_p^2(t) - \rho_p^2(0)], \quad (\text{II-91})$$

The initial values are given by

$$\rho_p(0) = \rho_{p1} = \rho_p - \beta$$

and

$$\dot{\rho}_p(0) = \gamma p^0. \quad (\text{II-92})$$

Inserting the initial values of Eqs. (II.92) and also inserting Eq. (II-88a) results in the following forms for the first integral:

$$\Lambda [\dot{\rho}_p(t) - \gamma p^0] = \frac{1}{2} [\rho_p^2(t) - \rho_{p1}^2] \quad (\text{II-93a})$$

and

$$\Lambda \gamma [p(t) - p^0] = \frac{1}{2} [\rho_p^2(t) - \rho_{p1}^2] \quad (\text{II-93b})$$

As in the case of the differential equation, the first integral is investigated at the time of the maximum of the transient. Application of Eq. (II.93b) to the maximum, where $\rho_p = 0$, yields:

$$\Lambda \gamma (p_m - p^0) = -\frac{\rho_{p1}^2}{2}. \quad (\text{II-94})$$

This gives a formula for the maximum flux amplitude or power of the excursion:

$$p_m = p^0 - \frac{\rho_{p1}^2}{2\Lambda\gamma}. \quad (\text{II-95})$$

The maximum power rise in megawatts with an energy coefficient γ_e [\$/MW · s] is

$$P_m - P^0 = -\frac{\rho_{p1}^2}{2\Lambda\gamma_e}. \quad (\text{II-96})$$

Note that the power rise $P_m - P^0$ is inversely proportional to the generation time and the energy coefficient, and proportional to ρ_{p1}^2 . But it is independent of the initial power if the temperature dependence of the energy coefficient is neglected.

As an example, again inserting the power reactor feedback coefficient of Eq. (II.47) yields:

$$\begin{aligned} \rho_1 &= 1.1\% ; \quad \beta/\Lambda = 0.62 \times 10^4 \text{ s}^{-1}; \quad \gamma_e = -0.8 \text{ } \$/P_n \text{ s} \\ P_m - P^0 &= \frac{1}{2} (0.1)^2 0.62 \times 10^4 \text{ s}^{-1} \cdot \frac{1}{0.8} P_n \text{ s} = 38.8 P_n \end{aligned} \quad (\text{II.97a})$$

Using $P^0 = 11P_0$ from Eq. (II-75) gives:

$$\begin{aligned} P_m &= 49.8 P_n \\ &\approx 1.25 \times 10^5 \text{ MW for } P_n = P_0 = 2500 \text{ MW} \end{aligned} \quad (\text{II.97b})$$

for a transient starting at the nominal power P_n . If the transient starts at a low power, e.g., $P_0 = 0.1 P_n$, then $P_m - P^0 = 388 P_0$ and P_n in the example becomes equal to $399 P_0$.

In the example, the rate of reactivity change at the maximum power, given by Eq. (II-86), is

$$\dot{\rho}_m = -\frac{0.8}{P_n} \$/s \cdot 49.8 P_n = -40 \$/s. \quad (\text{II.97c})$$

The first integral, Eq. (II.91), also provides information on the *total* reactivity feedback during the transient. Consider, for example, the transient terminated when $p(t)$ has returned to the pseudo-initial flux, p^0 , say at $t = t_2$. Then the left side of Eq. (II.93b) vanishes initially as well as when $p(t)$ is passing through p^0 at t_2 , i.e., at the "end" of the transient. Thus,

$$\rho_p^2(t) = \rho_{p1}^2 \text{ for } t = 0 \text{ and } t_2,$$

with the two solutions

$$\rho_p(0) = +\rho_{p1} \text{ at } t = 0$$

and

$$\rho_p(t_2) = -\rho_{p1} \text{ at } t = t_2. \quad (\text{II.98})$$

Thus, *the prompt feedback compensates the prompt reactivity twice during the transient* (see Fig. II-5). In other words, the reactivity swing is just two times the initial prompt reactivity.

Equations (II.98), (II-76), and (II-83) show that the total energy release during a superprompt-critical transient (i.e., up to t_2) is just twice the energy release up to the maximum of the transient:

$$\Delta Q(t_2) = 2\Delta Q_m = -2 \frac{\rho_{p1}}{\gamma_e}. \quad (\text{II.99})$$

Thus, *the total energy release during a superprompt-critical transient is independent of the generation time Λ* , as it is for delayed supercritical transients.

In the $\rho_1 = 1.1\%$ example, the average fuel temperature during the superprompt-critical transient is increased by 10°C or 100°C, respectively, assuming the temperature coefficients of Eqs. (II.46).

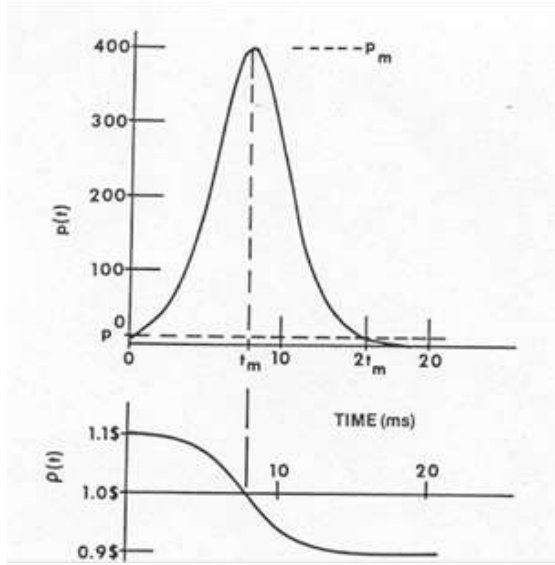


Figure II-5. Superprompt-critical transient ($\rho_1 = 1.1\%$, $p_0 = 0.1p_n$) from the PKA with linear energy feedback model.

Flux Transient During a Superprompt-Critical Excursion

The first integral in the form of Eq. (II.93a) is a first-order differential equation for

$\rho_p(t)$ that may be solved readily. Then $p(t)$ can be derived from $\rho_p(t)$. Rearranging the first integral, Eq. (II.93a) yields

$$2\Lambda \dot{\rho}_p(t) = \rho_p^2 - \rho_b^2, \quad (\text{II.100a})$$

with

$$\rho_b^2 = \rho_{p1}^2 - 2\Lambda \gamma p^0 > \rho_{p1}^2 \quad (\text{II.100b})$$

Dividing Eq. (II.100a) by its right side and integrating with respect to time from 0 to t yields

$$t = 2\Lambda \int_{\rho_{p1}}^{\rho_p(t)} \frac{d\rho_p}{\rho_p^2 - \rho_b^2} = 2\Lambda \int_{\rho_p(t)}^{\rho_{p1}} \frac{d\rho_p}{\rho_b^2 - \rho_p^2} \quad (\text{II.101})$$

or

$$\frac{t}{\Lambda} = \frac{1}{\rho_b} \left[\ln \frac{\rho_b + \rho_{p1}}{\rho_b - \rho_{p1}} - \ln \frac{\rho_b + \rho_p(t)}{\rho_b - \rho_p(t)} \right]. \quad (\text{II.102})$$

By setting $\rho_p(t)$ in Eq. (II.102) equal to zero, one obtains the time, t_m , at which the flux maximum of the transient occurs:

$$\frac{t_m}{\Lambda} = \frac{1}{\rho_b} \ln \frac{\rho_b + \rho_{p1}}{\rho_b - \rho_{p1}}. \quad (\text{II.103})$$

Equation (II.103) may be used to simplify Eq. (II.102), which can then be written in the form

$$\frac{t - t_m}{\Lambda} = -\frac{1}{\rho_b} \ln \frac{\rho_b + \rho_p(t)}{\rho_b - \rho_p(t)}. \quad (\text{II.104})$$

The desired solution, i.e., the t dependence of ρ_p , is obtained by solving Eq. (II.104) for $\rho_p(t)$. This gives, as the *solution of the differential equation*, Eq. (II-89):

$$\rho_p(t) = \rho_b \frac{1 - \exp\left[\frac{\rho_b}{\Lambda}(t - t_m)\right]}{1 + \exp\left[\frac{\rho_b}{\Lambda}(t - t_m)\right]}. \quad (\text{II.105})$$

According to this equation, the prompt reactivity is reduced from its initial $\rho_p(0) = \rho_{p1}$ to zero at t_m and to $-\rho_{p1}$ at $t_2 = -2t_m$. It can also be shown that $\rho_p(t)$ is antisymmetric around $t = t_m$ [compare Fig. II-5 for $\rho(t)$ in a transient induced $\rho_1 = 1.1\%$].

Inserting Eq. (II.105) into Eq. (II.93b), using the definition of ρ_b of Eq. (II.100b) and the value of the maximum flux p_m of Eq. (II.95), i.e.,

$$p_m = -\frac{\rho_b^2}{2\Lambda\gamma}, \quad (\text{II.106})$$

yields for the flux transient:

$$p(t) = \frac{p_m}{\left\{ \cosh\left[\frac{\rho_b}{2\Lambda}(t - t_m)\right] \right\}^2}. \quad (\text{II.107})$$

Thus, the PKA together with the linear energy feedback model gives a transient that is symmetrical around $(t - t_m)$. Figure II-5 shows the flux transient described by Eq. (II.107) for a transient induced by a step $\rho_1 = 1.1\%$, $p_0 = 0.1p_n$, and $\beta/\Lambda = 0.62 \times 10^4 \text{ s}^{-1}$.

The simplest formula for the "width" of the flux pulse (or burst) is obtained when the width $\bar{\Delta t}$ is defined by the flux integral divided by the maximum value:

$$\bar{\Delta t} = \frac{1}{p_m} \int_0^{2t_m} p(t) dt = -\frac{2}{p_m} \frac{\rho_{p1}}{\gamma}. \quad (\text{II.108})$$

The flux integral in Eq. (II.108),

$$\int_0^{2t_m} p(t) dt = -2 \frac{\rho_{p1}}{\gamma}, \quad (\text{II.109})$$

follows directly from Eq. (II.99). The maximum flux, p_m , is given by Eq. (II.94). By using the identity,

$$\frac{1}{p_m} = \frac{1}{p_m - p^0} \left(1 - \frac{p^0}{p_m} \right),$$

and Eq. (II.94), $\bar{\Delta t}$ is obtained as

$$\bar{\Delta}t = \frac{4\Lambda}{\rho_{p1}} \left(1 - \frac{p^0}{p_m} \right). \quad (\text{II.110})$$

If p_m is much greater than the pseudo-initial flux p^0 , the width becomes equal to four initial prompt periods:

$$\bar{\Delta}t = \frac{4\Lambda}{\rho_{p1}} \quad \text{for } p_m \gg p^0. \quad (\text{II.111})$$

The proportionality of $\bar{\Delta}t$ to Λ compensates the $1/\Lambda$ dependence of p_m and thus yields Λ - independent energy production. Figure II-6 illustrates the definition, Eq. (II.108), of the width of the power burst ($\rho_{p1} = 0.1\%$, and $\beta/\Lambda = 0.62 \times 10^4 \text{ s}^{-1}$, $\bar{\Delta}t \approx 6.5 \text{ ms}$).

Figure II-6 also shows that t_m , the time to reach the maximum, is not much larger than the burst width if the transient starts at the nominal power. The maximum is quickly reached after the onset of the burst. The value of t_m can be derived for a small initial power by simplifying Eq. (II.103):

$$t_m \approx \frac{\Lambda}{\rho_{p1}} \left(\ln 4 \frac{p_m}{p^0} \right) \quad \text{for } p^0 \ll p_m. \quad (\text{II.112})$$

If P_0 is 10^{-6} times the nominal power, then the example of Eqs. (II.97) gives

$$t_m \approx 16.5 \frac{\Lambda}{\rho_{p1}} \approx 4\bar{\Delta}t. \quad (\text{II.113})$$

That is, even for very low initial power, the time to reach the maximum in a step-induced transient is only several times the width of the power burst.

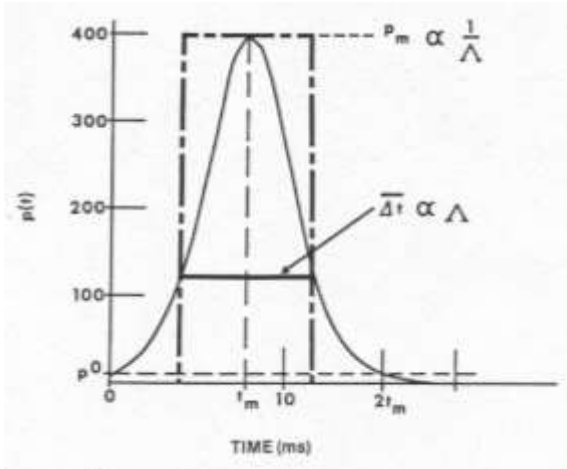


Figure II-6. Illustration of the width, Eq. (II.108), of the superprompt-critical transient presented in Fig. II-5.

Post-Burst Flux Transient

The neutron flux as described by Eq. (II.107) decreases rapidly after it passes through its maximum value (see also Fig. II-5). Formally, this rapid die-away continues beyond $t_2 = 2t_m$, the time that was used above as the "end" of the flux pulse. This further decrease of the flux and its asymptotic approach to zero is physically unrealistic; the reactor is still supercritical (if ρ_1 is smaller than 2β , which should always apply in reactor problems). The reason for this unrealistic behavior of the solution of Eq. (II.107) is the neglect of the delayed neutron source except as a modification of the initial condition.

At $t = t_m$, the neutron flux level is almost completely determined by the large number of neutrons that were produced during the time the reactor was superprompt critical. After the reactivity is reduced below β , the neutron flux of the superprompt reactor dies away rapidly, and the flux is eventually determined by the source multiplication of the delayed neutrons. The continuing die-away as it follows from the PKA then becomes totally unrealistic.

After the rapid flux changes during the burst are terminated, the further changes of the flux are slow so that $\Lambda \dot{p}$ can be neglected (prompt jump approximation). An estimate of the flux level can then be obtained by estimating the delayed neutron source and applying the time-dependent source multiplication formula. A first approximation is obtained by using the initial value of the delayed neutron source and the reactivity after the burst, i.e.,

$$\rho(t_2) = \beta - \rho_{p1}. \quad (\text{II.114})$$

From the source multiplication consideration of the delayed neutrons, it follows that the flux is “held up” at least at a level p_{ab} (after the burst) given by:

$$p_{ab} \approx \frac{s_{d0}}{\beta - \rho} = \frac{s_{d0}}{\rho_{p1}} = \frac{\beta}{\rho_{p1}} p_0. \quad (\text{II.115})$$

Thus, in the example above, the flux is held up at a level of ten times the initial power.

An improved estimate of the delayed neutron source multiplication requires accounting for the increase of the precursor population during the flux burst. During the short time of the burst, the $\bar{\lambda}$ kinetics approximation is applicable. Since the time interval considered is very small, the precursor decay can be neglected in the treatment of the precursor balance ($\bar{\lambda} t \ll 1$). Thus, $\zeta(t_2)$ can be approximated by:

$$\zeta(t_2) \approx \zeta_0 + \beta \int_0^{t_2} p(t') dt'. \quad (\text{II.116})$$

with $\bar{\lambda} \zeta_0 = \beta p_0$, and the flux integral given by Eq. (II.109), $\bar{\lambda} \zeta$ is obtained as

$$\bar{\lambda} \zeta(t_2) = \beta p_0 \left(1 - 2 \frac{\rho_{p1} \bar{\lambda}}{p_0 \gamma} \right). \quad (\text{II.117})$$

In the example treated above, i.e., $p_0 = 0.1 p_n$, the delayed neutron source is increased by ~140%:

$$-2 \frac{\rho_{p1} \bar{\lambda}}{p_0 \gamma} = \frac{-2(0.1\$)(0.565/s)}{(0.1)(-0.8\$/s)} \approx 1.41. \quad (\text{II.118})$$

If the initial power was very low, the fraction of the precursors produced during the flux burst would be even larger.

Instead of Eq. (II.115), the flux after the burst with the precursor increase included is given by:

$$p_{ab} \approx \frac{\beta p_0}{\rho_{p1}} \left(1 - 2 \frac{\rho_{p1} \bar{\lambda}}{p_0 \gamma} \right). \quad (\text{II.119})$$

Figure II-7 shows the flux according to Eq. (II.107) (dashed line) and the result of the $\bar{\lambda}$ kinetics model (solid line). The six-delay-group results are indistinguishable from the one-delay-group results at the small times of Fig. II-7. The delayed neutron source holds the flux at $p_{ab} \approx 2.4/0.1$ in accordance with Eq.

(II.119). The deviations between the prompt kinetics approximation and the actual calculated results during and after the burst, shown in Fig. II-7, give a good indication of the applicability of the prompt jump approximation.

The further progression of the excursion is essentially the same as in a transient following a subprompt-critical step reactivity insertion. The reactor power after the prompt burst is much larger than the nominal power; this causes a further increase in the fuel temperature and a corresponding decrease in the reactivity due to the negative energy coefficient. Eventually, the reactivity is reduced to zero and the flux approaches an asymptotic level above the initial flux.

This further decrease of the flux is indicated in Fig. II-7. Compared to Fig. II-4, the flux decrease in Fig. II-7 appears very slow due to the strongly expanded scale.

The temperature increase during the entire transient is composed of the part produced during the prompt burst and the part produced during the subsequent relatively gradual power decrease:

$$\Delta T_{total} = \Delta T_{burst} + \Delta T_{post-burst}$$

and

$$\Delta T_{total} = -\frac{1}{\gamma_T} [2(\rho_1 - \beta)] - \frac{1}{\gamma_T} [\rho_1 - 2(\rho_1 - \beta)] = -\frac{\rho_1}{\gamma_T} \quad (II.120)$$

In reactor accident analysis, ρ_1 is almost never much larger than β . Then, only the small part of the temperature rise occurs during the prompt burst; the larger part of the energy is released during the subprompt-critical phase of the excursion, before or after a prompt burst. For example, for $\rho_1 = 1.1\beta$, ~ 18% of the energy is released during the prompt burst and ~72% during the slower phase of the transient.

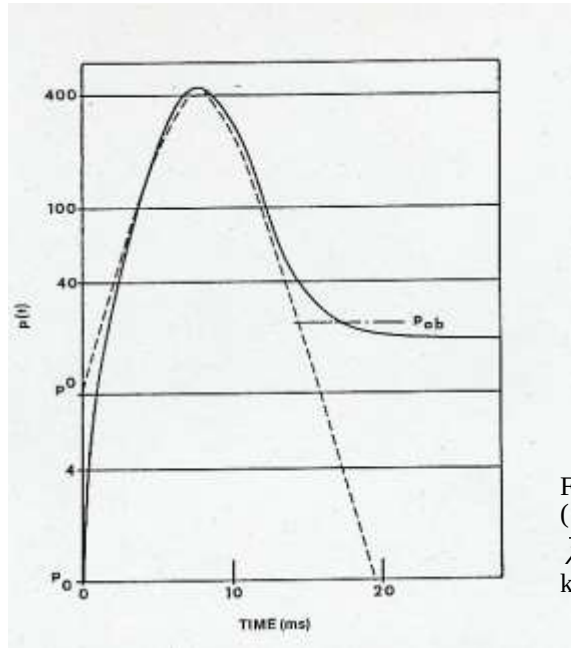


Figure II-7. Superprompt-critical transient ($\rho_1 = 1.1\beta$, $p_0 = 0.1p_n$) with reactivity feedback: λ kinetics model (solid line) and prompt kinetics of Eq. (II.107) (dashed line).

II-4 Superprompt-Critical Transients Induced by Reactivity Ramps

Investigation of the Differential Equation

Reactivity steps in the superprompt reactivity domain of reactors are only an idealization. Realistic simulation of superprompt-critical reactivity insertions in a reactor must account for the reactivity insertion rate, and thus for the rapid flux response that occurs *during* the reactivity insertion (and not *after* the reactivity insertion as in the case of step approximations). Therefore, this section investigates more realistic, gradual reactivity insertions, though in the idealized form of a continuing uninterminated ramp. In fully realistic situations, ramps are always terminated at some maximum available reactivity.

The PKA for reactivity ramps is applied (as for the step-induced transients) only after the time, t_p , when the reactor became superprompt critical:

$$\dot{p}(\tau) = \frac{\rho_p(\tau)}{\Lambda} p(\tau) \quad \text{with } \tau = t - t_p. \quad (\text{II.121})$$

which is solved with $p(0) = p^0$, and $\rho_p(0) = 0$ for $\tau = 0$. For reactivity ramps, the pseudo-initial flux, p^0 , is approximately two times the actual flux at prompt critical:

$$p^0 \approx 2p_{pc} \approx p_0 \beta \left(\frac{2\pi}{\Lambda a} \right)^{1/2}. \quad (\text{II.122})$$

The prompt reactivity, with the adiabatic approximation applied to the energy above p_0 is given by

$$\rho_p(\tau) = a\tau + \gamma \int_0^\tau [p(\tau') - p_0] d\tau' \quad (\text{II.123a})$$

or

$$\rho_p(\tau) = a'\tau + \gamma \int_0^\tau p(\tau') d\tau', \quad (\text{II.123b})$$

with

$$a' = a + \gamma p_0. \quad (\text{II.123c})$$

The stationary cooling described by the term " $-p_0$ " in the integrand of Eq. (II.123a) is now included in the feedback reactivity since this term only modifies the ramp rate, Eq. (II.123c). In the case of step-induced transients, stationary cooling was neglected for mathematical simplification.

Initial results about essential characteristics of the transients are deduced directly from the differential equation as it was done for the step-induced transient. It follows from Eq. (II.121) that $\dot{p}(\tau)$ and $\rho_p(\tau)$ pass through zero simultaneously, i.e.,

$$\dot{p}(\tau) = 0 \quad \text{when} \quad \rho_p(\tau) = 0.$$

Since $\rho_p(\tau)$ is zero initially, $p(\tau)$ in the PKA starts with a zero slope and has a *minimum* at $\tau = 0$. Subsequently, $\rho_p(\tau)$ and $p(\tau)$ both increase. With increasing accumulation of energy,

the reactivity feedback overcomes the ramp reactivity and $\rho_p(\tau)$ is again reduced to zero; say at $\tau = \tau_{m1}$. The flux has its first maximum at this time. After passing through zero, ρ_p approaches a negative value, corresponding to $-\rho_{p1}$ in the case of a step-induced transient. By this time, the high flux during the prompt burst is reduced so that further reactivity reduction through feedback occurs relatively slowly (compare Fig. II-5). A similar reduction of the reactivity is to be expected in the case of a ramp reactivity insertion. However, as the ramp continues to increase the reactivity, it again becomes equal to β . At this time, a second flux minimum occurs and the cycle repeats; see Fig. II-8 where $p_0 = p_n$, $a = 50\$/s$, $\gamma = -0.5\$/\beta p - s$, and $p^0 = 28.0 p_0$ according to Eq. (II.122).

The discussion of maxima and minima showed that the ramp-induced transient treated in the PKA consists of a repetition of superprompt flux bursts. The maxima and minima of the flux occur when $\rho(t)$ passes through β :

$$p(t) = p_{\max} \text{ or } p_{\min} \text{ for } \rho_p(t_{ml}) = 0. \quad (\text{II.124})$$

The succession of bursts can be characterized by an averaged power that can be determined as follows: Let Δt_l be the duration of the l^{th} burst and ΔQ_l the corresponding energy release:

$$\Delta Q_l = \int_{\Delta t_l} [P(t') - P_0] dt'. \quad (\text{II.125})$$

Since $\rho_p(t)$ is zero at the beginning and at the end of each burst, a relation between the reactivity insertion and the feedback reactivity during a burst can be obtained:

$$0 = a\Delta t_l + \gamma_e \int_{\Delta t_l} [P(t') - P_0] dt' = a\Delta t_l + \gamma_e \Delta Q_l. \quad (\text{II.126})$$

The energy release per burst time, i.e., the "average" power increase $\Delta \bar{P}$ during the burst, is obtained from Eq. (II.126) as:

$$\frac{\Delta Q_l}{\Delta t_l} = \Delta \bar{P} = \bar{P} - P_0 = -\frac{a}{\gamma_e}. \quad (\text{II.127a})$$

The average power $\Delta \bar{P}$ is independent of the generation time and of the initial power; it is the same for all bursts under the conditions of the feedback model. Equation (II.127a) can be rewritten for the flux amplitude by using the effective ramp rate a' of Eq. (II.123c):

$$\bar{p} = -\frac{a'}{\gamma}. \quad (\text{II.127b})$$

The horizontal dashed line in Fig. II-8 represents $\bar{p}$. For this transient, Eqs. (II.127) yield:

$$\Delta \bar{p} = 100 p_n$$

and

$$\bar{p} = 101 p_n. \quad (\text{II.128})$$

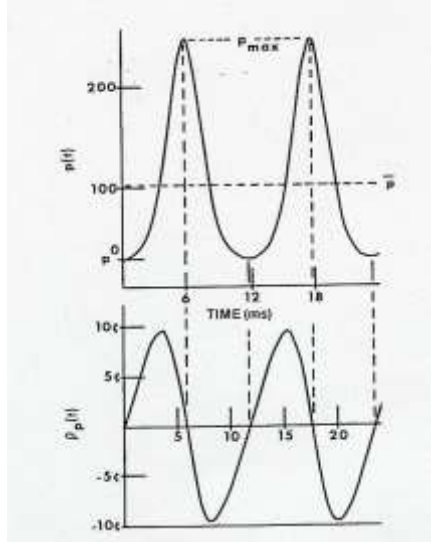


Figure II-8. The flux, $p(t)$, and the prompt reactivity $\rho_p(t)$, are shown for a reactivity-ramp-induced transient with temperature-independent energy coefficient where $p_0 = p_n$, $a = 50 \$/s$, $\gamma = -0.5 \$/fp - s$, and $p^0 = 28.0 p_0$.

Note, for a *step-induced* supercritical transient, information on the *energy* release was obtained by investigating the transition of ρ through β ; for a *ramp-induced* transient, information on the *average power* is obtained instead. This difference occurs because only the *rate* of reactivity insertion is given for ramp-induced transients, which then determines the *rate* of energy release:

$$-\frac{\rho_{p1}}{\gamma_e} \Rightarrow \text{energy, for reactivity steps} \quad (\text{II.129a})$$

and

$$-\frac{a}{\gamma_e} \Rightarrow \text{average power, for reactivity steps.} \quad (\text{II.129b})$$

Additional information is obtained by investigating the extrema of $\rho_p(\tau)$. Setting the time derivative of Eqs. (II.123) equal to zero gives

$$\dot{\rho}_p(\tau) = a + \gamma(p - p_0) = a' + \gamma p = 0. \quad (\text{II.130})$$

From Eq. (II.130), it follows that the flux at the extrema of the reactivity is equal to the average flux, $\bar{p}$:

$$p(\rho_p^{\text{max,min}}) = p_0 - \frac{a}{\gamma} = -\frac{a'}{\gamma} = \bar{p}. \quad (\text{II.131})$$

i.e., the extrema of the reactivity coincide with $p(t)$ passing through $\bar{p}$ (see Fig. II-8).

The extrema of the reactivity determine the inflection points of the flux transient in a semi-logarithmic presentation:

$$\frac{1}{\Lambda} \frac{d}{d\tau} \rho_p(\tau) = \frac{d}{d\tau} \frac{\dot{p}}{p} = \frac{d^2}{d\tau^2} \ln p(\tau). \quad (\text{II.132})$$

Thus, the second derivative of the logarithm of the flux is zero at the extrema of the reactivity.

Investigation of the First Integral

The integration of the kinetics equation, Eq. (II.121), is facilitated by multiplying it with $\dot{\rho}_p$; on the left side, $\dot{\rho}_p$ is expressed in terms of p as given by Eq. (II.130):

$$(a' + \gamma p) \dot{p} = \frac{1}{\Lambda} \rho_p \dot{\rho}_p p, \quad (\text{II.133})$$

with a' given by Eq. (II.123c). After dividing Eq. (II.133) by p , the resulting equation may be readily integrated:

$$\int_0^\tau \left(\frac{a'}{p} + \gamma \right) \dot{p} d\tau' = \frac{1}{\Lambda} \int_0^\tau \rho_p \dot{\rho}_p d\tau'$$

The derivatives of p and ρ_p may be conveniently combined with $d\tau'$. Then, the two integrals are carried out over p and ρ_p , respectively:

$$\int_{p(0)}^{p(\tau)} \left(\frac{a'}{p} + \gamma \right) dp = \frac{1}{\Lambda} \int_{\rho_p(0)}^{\rho_p(\tau)} \rho_p d\rho_p.$$

The integrations may be readily carried out. Inserting $p(0) = p^0$ and $\rho_p(0) = 0$ as the lower limits gives the "first integral," which represents a relation between the flux and the reactivity:

$$a' \ln \frac{p}{p^0} + \gamma(p - p^0) = \frac{1}{2\Lambda} \rho_p^2. \quad (\text{II.134})$$

The first integral is used to find the maximum and minimum values of both the reactivity and the flux transients (compare Fig. II-8). Inserting $\bar{p}$ from Eq. (II.131) into Eq. (II.134) and using Eq. (II.131) to simplify the result yields the maximum and minimum values of ρ_p :

$$(\rho_p^{\max, \min})^2 = 2\Lambda a' \left(\ln \frac{\bar{p}}{p^0} - 1 + \frac{p^0}{\bar{p}} \right), \quad (\text{II.135})$$

which gives the two values

$$\frac{\rho_p^{\max, \min}}{\beta} = \pm \frac{\sqrt{2\Lambda a'}}{\beta} \sqrt{\ln \frac{\bar{p}}{p^0} - 1 + \frac{p^0}{\bar{p}}}, \quad (\text{II.136})$$

The positive and negative roots represent the reactivity maxima and minima, respectively. Apparently, the maxima and minima of ρ are located symmetrically about β .

Maximum reactivity values, calculated from Eq. (II.136) for $a = 50$ \$/s ($\Lambda a' = 2.49 \times 10^{-5}$) are plotted in Fig. II-9 as a function of $\bar{p} / p^0$. The abscissa in Fig. II-9 covers all transients with realistic ramp rates and feedback coefficients starting at (or around) nominal power. The lower end of the scale is $\bar{p} / p^0 = 1$ since Eq. (II.136) has real roots only if $\bar{p} / p^0 \geq 1$. If Eq. (II.136) has no real roots, i.e., if

$$\bar{p} < p^0, \quad (\text{II.137})$$

with $\bar{p}$ from Eq. (II.127b), then the PKA is fundamentally inapplicable. The inequality (II.137) may be physically realized for small ramp rates, large feedback coefficients (small $\bar{p}$), or a large pseudo-initial flux (large p^0). In such cases, the feedback effect during the subprompt-critical part of the transient is strong enough that the reactivity does not reach the superprompt-critical domain. The upper end of the scale in Fig. II-9 may be exceeded only at very low starting power (e.g., accidents during reloading of the core).

Figure II-9 shows that a ramp rate of 50\$/s can only push a fast reactor beyond prompt critical by a small fraction of a dollar. Results for different ramp rates may be obtained readily from Fig. II-9 by changing the factor in front of the square root of Eq. (II.136).

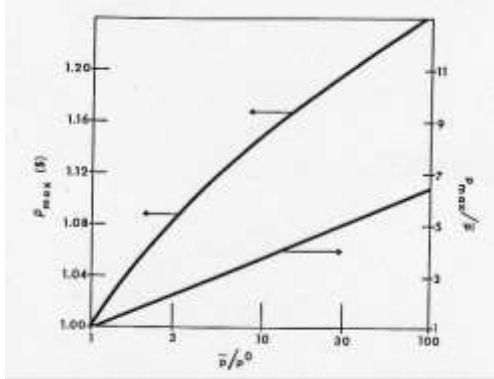


Figure II-9. Maximum reactivity values, Eq. (II.136), and $\rho_{\max} / \bar{p}$, Eq. (II.138), versus $\bar{p} / p^0$ for a reactivity-ramp-induced transient where $a = 50$ \$/s and $\gamma = -0.5$ \$/fp-s.

For example, an increase of the ramp rate from 50 to 100\$/s increases the maximum of ρ_p by a factor of ~ 1.4 . Since positive ramp rates beyond 100\$/s are very unlikely, it follows that realistic ramp rates can drive the reactivity in a fast reactor by only a fraction of a dollar beyond prompt critical. Much higher reactivity values are attainable, in principle, in thermal reactors due to the larger generation time. However, thermal reactors do not have the positive reactivity insertion mechanisms of fast reactors, as they may result from coolant boiling or fuel motion.

When the prompt reactivity passes through zero, the flux has a maximum or a minimum, $p = p_m$. Setting $\rho_p = 0$ and $p = p_m$ in the first integral, Eq. (II.134), yields a transcendental equation for the flux maxima and minima (if properly expanded, $\bar{p} / p^0$ appears as the only parameter):

$$\frac{\bar{p}}{p^0} \ln \left(\frac{p_m}{\bar{p}} \frac{\bar{p}}{p^0} \right) - \frac{p_m}{\bar{p}} \frac{\bar{p}}{p^0} + 1 = 0, \quad (\text{II.138})$$

which is satisfied for $p_m = p^0$ and gives the flux minima as p^0 :

$$p_{\min} = p^0. \quad (\text{II.139})$$

The flux maxima are given by the second positive solution of Eq. (II.138).

The ratio of the flux maxima to the average flux, i.e., $p_{\max} / \bar{p}$, is also plotted in Fig. II-9 as a function of $\bar{p} / p^0$. The figure shows that $p_{\max}$ does not become very much larger than $\bar{p}$. Although the ramp rate does not explicitly appear in Eq. (II.138), $p_{\max} / \bar{p}$ increases with increasing ramp rate since $\bar{p} / p^0$, is approximately proportional to $(a)^{3/2}$.

The application of Fig. II-9 is illustrated by deriving results for the transient presented in Fig. II-8. With $p_0 = p_n$, $p^0 = 28p_0$, and $\bar{p} = 101p_n$, then $\bar{p}/p^0 = 3.6$. At $\bar{p}/p^0 = 3.6$, Fig. II-9 yields

$$\frac{p_{\max}}{\bar{p}} \approx 2.4 \text{ and } \rho_p^{\max} = 9.5\%$$

Both values are in complete agreement with the results of Fig. II-8.

The investigation of the first integral for ramp-induced transients provided important information such as $\bar{p}$, $\rho_{\max}$, and $p_{\max}$. The latter two quantities are depicted in Fig. II-9 as functions of $\bar{p}/p^0$ while $\bar{p}$ is given by Eq. (II.131) and p^0 may be obtained, approximately, from Eq. (II.122). The two quantities, $\bar{p}$ and p^0 , are shown in Fig. II-10 as functions of the ramp rate. The pseudo-initial flux p^0 decreases and the average flux increases with increasing ramp rate. The PKA can only be applied for ramp rates larger than the crossover point (a^*) of the $p^0(a)$ and $\bar{p}(a)$ curves. If the reactivity insertion rate is smaller than a^* , the transient is comparatively slow. Thus, enough time is allowed to accumulate the energy, and thus the feedback reactivity, that is needed to prevent the reactor from becoming superprompt critical.

Figure II-10 shows that in the domain of applicability of the PKA, the simple formulas for p^0 and $\bar{p}$,

$$p^0 = p_0 \beta \sqrt{\frac{2\pi}{\Lambda a}} \quad (\text{II.140a})$$

and

$$\bar{p} = -\frac{a'}{\gamma}, \quad (\text{II.140b})$$

which are presented by the straight dashed lines, are sufficiently accurate for semi-quantitative estimates.

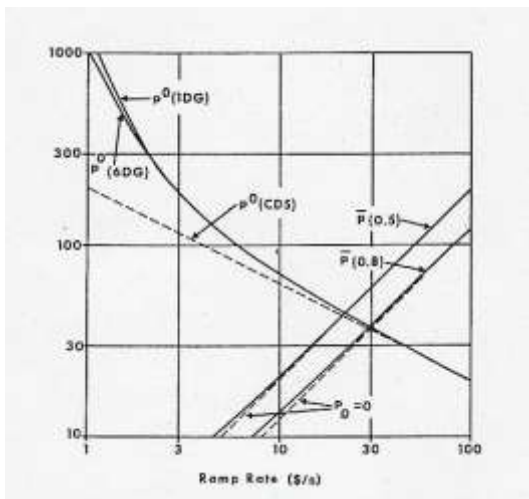


Figure II-10. For various approximations, p^0 and $\bar{p}$ ($\gamma = -0.5$ and -0.8 \$/fp-s) versus reactivity ramp rate; the dashed lines are from Eqs. (II.140).

Discussion of the Flux Transient

The calculation of the flux as a function of time requires the complete solution of the differential equation (II.121). An analytical solution of this equation with the reactivity from Eqs. (II.123) has not been found yet. It is, however, possible to calculate the time between the bursts and then deduce the energy per burst from the ratio of the average power, $\bar{P}$, Eq. (II.127a), and the duration of a burst. The calculation of the burst duration, however, for realistically mild transients is very complicated. Therefore, the simple solution of the step-induced transients with a proper ρ_{p1} value is frequently related to ramp-induced transients.

The similarity of ramp- and step-induced superprompt-critical bursts was observed experimentally. Theoretical relations for ramp- and step-induced bursts are based on matching the values of the burst maxima; i.e., p_m of Eq. (II.138) is set equal to p_m of Eq. (II.95):

$$[p_m(a), \text{ from Eq. (II.138)}] = [p_m(\rho_{p1}) \text{ from Eq. (II.95)}] \quad (\text{II.141})$$

The resulting equivalence relation is relatively simple but only because the ramp rate was assumed to be very large and thus ρ_1 is much larger than β . The investigations in the previous sections showed that this assumption is not realistic for large fast power reactors; it may, in principle, be realized in thermal reactors.

An equivalence relation with a wider range of applicability can be based on the temperature rise during the burst. It applies the formula for the energy production per burst. The equivalence relation presented here is based on the maximum reactivity that can be obtained from the first integral:

$$[\rho_p^{\max}(a), \text{ from Eq. (II.136)}] = [\rho_{p1} \text{ for steps}] \quad (\text{II.142})$$

The equivalence relation, Eq. (II.142), makes it possible to apply the simple formulas for step-induced transients to the much more complicated ramp-induced bursts. Figure II-11 shows the maximum reactivity calculated from Eqs. (II.136) and (II.140) as a function of the ramp rate. Since $\rho_{p,\max}$ is set equal to ρ_{p1} , Fig. II-11 yields directly the ramp rate equivalent to the step reactivity ρ_{p1} .

The previous investigations in this section as well as the numerical solution presented in Fig. II-8 are based on the PKA; i.e., the delayed neutron source is omitted and the resulting error is approximately corrected by starting the transient from a pseudo-initial flux, p^0 . The value of p^0 is determined by matching the PKA solution with a more correct result on the rising wing of the first burst. This adjustment yields a fairly accurate PKA solution during the major part of the burst. The comparison of the PKA for step-induced bursts and the corresponding numerical solution showed that the PKA becomes invalid "after" the burst, i.e., after the flux is reduced to a value comparable to the flux sustained by the delayed neutron source [see Fig. II-7 and Eq. (II.118)]. The same phenomenon is to be expected for ramp-induced bursts, i.e., the flux is *not* expected to decrease to the p^0 value as in Fig. II-8.

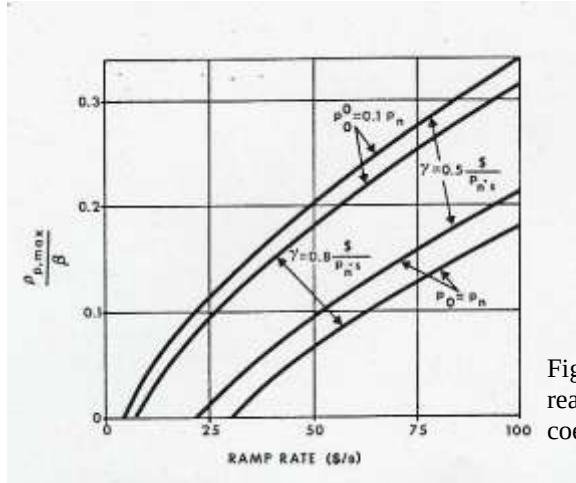


Figure II-11. The maximum reactivity versus reactivity ramp rate for two different energy coefficients and initial conditions.

Figure II-12 shows a comparison between the flux transient in the PKA (dashed line) with the actual solution. The striking difference is the occurrence of significant damping, which results from the proper treatment of the delayed neutrons. The important fact is that even though the oscillations are damped, the flux still oscillates around the *same* average value $\bar{\rho}$ as in the PKA. Because of the strong damping, the PKA for the burst shape is only applicable during the first burst.

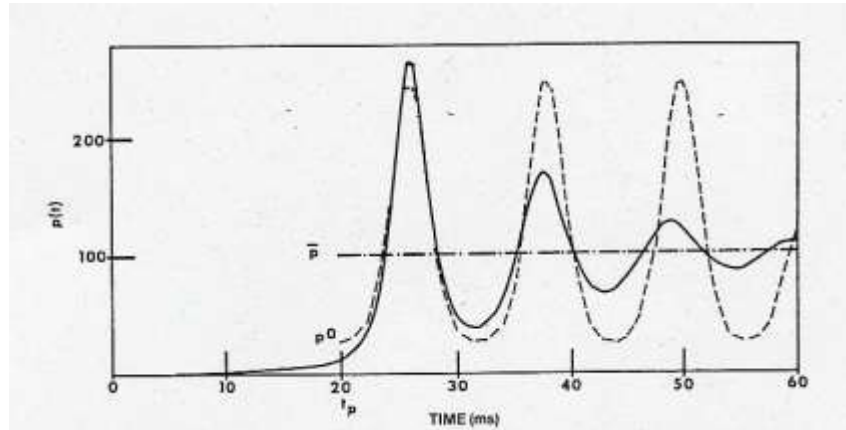


Figure II-12. The flux transient induced by a reactivity ramp in the PKA (dashed line) and the actual solution (solid line) with a temperature-independent energy coefficient.

During the first burst, the PKA yields fairly accurate results. The small deviation shown in Fig. II-12 is similar to that exhibited in Fig. II-7 for the step-induced transient. In both cases, the PKA flux in the initial phase of the transient is high due to the artificially high starting value, p^0 . The additional power produced during this phase results in an earlier reduction of the reactivity through feedback. The flux maxima in the PKA are consequently somewhat lower than the correct solution.

This situation is reversed for subsequent bursts in ramp-induced transients. The actual flux is higher than the PKA flux due to source multiplication of the delayed neutron source with a reactivity close to β . This results in a substantial reactivity feedback even during the phase in

which $p(t) < \bar{p}$. Therefore, subsequent bursts are driven by a smaller effective ramp rate. A further contribution to the damping effect of the delayed neutrons stems from the maximum flux being smaller if p^0 is larger, even for the same ramp rate; it follows from Fig. II-9 that $p_{\max}$ approaches $\bar{p}$ from above if p^0 approaches $\bar{p}$ from below.

The energy coefficient was assumed independent of the temperature in all of the solutions presented in this section. The results for single bursts are fairly insensitive to this assumption. Temperature dependence can be taken into account approximately by using a proper average value. In a succession of bursts, as it occurs with high ramp rate reactivity insertions, the temperature dependence of γ has to be taken into account more explicitly.

According to Eq. (II.42), the Doppler effect contribution to the energy coefficient has approximately a $1/\sqrt{T}$ or a $1/T$ temperature dependence for thermal and fast reactors, respectively. The temperature increase during the first burst, therefore, causes a reduction in the average energy coefficient effective during the second burst, etc. Reduction of the energy coefficient, γ , from burst to burst has two effects:

1. It increases $\bar{p} \approx -\frac{a'}{\gamma}$.
2. It increases the maximum flux, which is measured, for example, by $p_m / \bar{p}$. Figure II-9 shows, for any given p^0 value, that a larger $p_m / \bar{p}$ value is obtained if $\bar{p}$ is larger.

Since $\bar{p}$ increases with temperature, it follows that the flux oscillates around a $\bar{p}$ curve that increases with time. An increase in $p_m / \bar{p}$ means an increase in the amplitude of the oscillation, i.e., a partial compensation of the damping effect that appears in Fig. II-12.

Figure II-13 shows the same transient as in Fig. II-12 (50\$/s) with an energy coefficient proportional to $1/T$. The temperature increase in the constant power removal model is obtained from Eqs. (II.31) and (II.32):

$$\delta T(t) = T(t) - T_0 = C_{\pi} \int_0^t [p(t') - p_0] dt' = C_{\pi} I(t). \quad (\text{II.143})$$

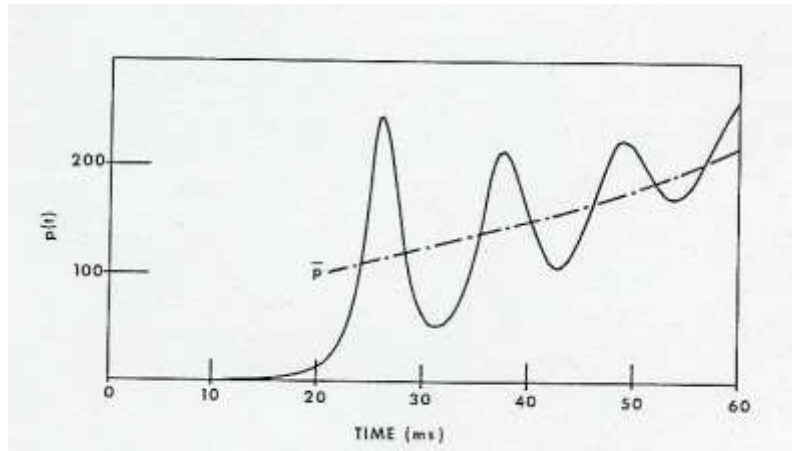


Figure II-13. The transient in Fig. II-12 with an energy coefficient proportional to $1/T$.

The reactivity feedback is found from Eq. (II.43) as:

$$\delta\rho(t) = \int_{T_0}^{T(t)} \frac{A_D}{T} dT = A_D \ln \frac{T(t)}{T_0}, \quad (\text{II.144})$$

or by inserting $T(t)$ from Eq. (II.143) as:

$$\delta\rho(t) = A_D \ln \left\{ \frac{T_0 + C_{IT} \int_0^t [p(t') - p_0] dt'}{T_0} \right\}. \quad (\text{II.145})$$

Eq. (II.145) replaces the simplified feedback reactivity calculated with a temperature-independent Doppler coefficient:

$$\delta\rho(t) = \gamma \int_0^t [p(t') - p_0] dt' = \gamma I(t). \quad (\text{II.146})$$

For a comparison of the two feedback reactivity, Eqs. (II.145) and (II.146), the contribution of expansion to the prompt feedback is neglected. (This neglect is frequently made in conservative safety calculations.) Then the values of γ and A_D in Eqs. (II.146) and (II.145), respectively, are directly related.

Dividing Eqs. (II.146) and (II.143) yields as the temperature coefficient (due to Doppler effect):

$$\frac{\delta\rho}{\delta T} = \left(\frac{d\rho}{dT} \right)_D = \frac{\gamma}{C_{IT}}. \quad (\text{II.147})$$

For comparison with the T -dependent Doppler coefficient, the right side of Eq. (II.147) is set equal to the average coefficient during the first burst:

$$\left(\frac{d\rho}{dT} \right)_D = \frac{\gamma}{C_{IT}} = \frac{A_D}{\bar{T}^{(1)}}. \quad (\text{II.148})$$

using the $1/T$ dependence of the fast reactor example presented here. The γ value from Fig. II-12 was used together with the C_{IT} value estimated in Eq. (II.35):

$$C_{IT} = 400 \text{ K/fp} \cdot \text{s}$$

and

$$\gamma = -0.5 \text{ } \$/\text{fp} \cdot \text{s}. \quad (\text{II.149})$$

In addition, a value for $\bar{T}^{(1)}$ is needed to estimate A_D from Eq. (II.148). The following value is used as a close approximation:

$$\bar{T}^{(1)} \approx T_0 + \frac{1}{2} \bar{\Delta T}^{(1)}. \quad (\text{II.150})$$

The temperature increase during the first burst, $\bar{\Delta T}^{(1)}$, in the example of Fig. II-12 ($T_0 = 1500 \text{ K}$), was estimated above as $\sim 200 \text{ K}$, which gives $\bar{\Delta T}^{(1)} \approx 1600 \text{ K}$. With these values, the following magnitude of A_D is obtained:

$$A_D = \frac{\gamma}{C_{IT}} \bar{T}^{(1)} = \frac{-0.5 \$/\text{fp-s}}{400 \text{ K}/\text{fp-s}} 1600 \text{ K} = -2\$. \quad (\text{II.151})$$

Using Eq. (II.151) yields, for the temperature coefficient for both approximations,

$$\gamma_T = \frac{A_D}{\bar{T}^{(1)}} = -1.25 \times 10^{-3} \frac{\$}{\text{K}} = -0.125 \frac{\$}{\text{K}}. \quad (\text{II.152})$$

If the temperature coefficient is constant, the same $\delta\rho$ is obtained for the same value of δT , independent of the temperature at which the increment δT is applied. For example, $\delta T = 160 \text{ K}$ yields $\delta\rho = 20\%$ independent of T .

For a $1/T$ -dependent coefficient, the same $\delta\rho$ is obtained for the same *relative* change of the temperature. Let T_2/T_1 be the ratio of the temperatures before and after the change ($T_2 = T_1 + \delta T$). The corresponding reactivity is given by

$$\Delta\rho = A_D \ln \frac{T_2}{T_0} = A_D \ln \frac{T_1}{T_0} + A_D \ln \frac{T_2}{T_1}. \quad (\text{II.153})$$

If, for example, $T_2/T_1 = 1.1$ ($\ln T_2/T_1 \approx 0.1$), then

$$\delta\rho = A_D \ln \frac{T_2}{T_1} \approx 0.1 A_D = 20\%.$$

The same value of 20% is obtained when T changes by 10% about any base temperature.

The transient depicted in Fig. II-13 clearly shows the expected features, i.e., oscillation of the flux around an increasing $\bar{\rho}$ (dashed-dotted line) and a lessened damping effect as compared to Fig. II-12.

The presentation of transients with feedback follows an inductive path: It starts with the most idealized reactivity insertion, the step, and the most idealized feedback, the adiabatic linear energy model, and proceeds by including (in well-defined increments) more and more reality in the transient treatment. Quantitative comparisons show the effect of the various changes in the physical model. Practical applications require detailed numerical solutions. However, the semi-quantitative feature developed here can serve as a useful guide toward setting up the problems and evaluating the numerical solutions.

Review Questions

1. Find the energy deposited in the fuel without and with heat transfer.
2. Why is the feedback reactivity approximated by the Nordheim-Hansen model (i.e., neglecting the stationary heat removal) in the analytical treatment of step-induced superprompt-critical transients? Questions 9 through 12 pertain to these transients.
3. Give the differential equation in the prompt kinetics approximation, including the initial condition.
4. Find the reactivity at the maximum of the transient.
5. Find the energy release up to the maximum of the transient.
6. Derive the formula for ρ at the maximum of the flux.
7. Transform the original equation for superprompt-critical transients with linear energy feedback into a second-order differential equation for the prompt reactivity ρ_p .
8. Find the first integral.
9. **a.** Find ρ_m the flux amplitude at the maximum.
b. Find the reactivity at $2t_m$.
10. Find the energy release at $2t_m$. Interpret the energy release and the corresponding temperature rise in terms of the respective reactivity coefficients.
11. Indicate the way the solution for $\rho_p(t)$ is derived from the first integral in a flux transient during a superprompt-critical excursion.
12. How do you find $p(t)$ from $\rho_p(t)$?
13. **a.** Define the width of the flux pulse.
b. Give the main factors of the formula for the width of the flux pulse and show why the energy release is independent of Λ .
14. **a.** Discuss the principal effect of the delayed neutron on the transient as treated in the prompt kinetics approximation.
b. Discuss the applicability of the prompt kinetics approximation (for $\rho > \beta$ and for $\rho < \beta$).
15. **a.** Estimate the power after the burst, say, at $t = 2t_m$:
b. Give the estimate without additional precursors.
c. How would the additional precursors that are produced during the burst affect this flux estimate?
d. Describe qualitatively the continuation of the transient after $t = 2t_m$.
16. **a.** Discuss the "shutdown capability" of the prompt negative reactivity feedback coefficients.
b. How does this "shutdown capability" depend on the magnitude of the energy coefficient?
17. Describe the "initial" condition as it is used in the prompt kinetics approximation for superprompt critical transients induced by reactivity ramps.

18. Give the differential equation, the initial condition, and the reactivity including prompt feedback.
19. Sketch qualitatively the time dependence of the reactivity and the flux in the prompt kinetics approximation and relate key points.
20. Find and discuss the information on energy release that can be extracted from the differential equation.
21. . Describe qualitatively the impact of the increase of the delayed neutron source on the first post-burst minimum.
 - b.** Explain the damping effect of the delayed neutrons.
 - c.** How do the delayed neutrons affect $\bar{P}$?
22. Suppose $\gamma(T)$ decreased with T ; describe the resulting two changes in the transient.